

BASIC NETWORK SNIFFER REPORT

CodeAlpha Cyber Security Internship

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1. Introduction

Network sniffing is an important concept in cyber security and networking. A network sniffer is a tool used to capture and monitor data packets traveling through a network. It helps security professionals analyze network traffic, troubleshoot network issues, and detect suspicious activities.

This project demonstrates a Basic Network Sniffer developed using Python and the Scapy library. The program captures packets and displays useful information such as the Source IP Address, Destination IP Address, and Protocol details.

2. Objective

The main objectives of this project are:

- Understand packet sniffing concepts.
 - Capture network packets using Python.
 - Analyze source and destination IP addresses.
 - Learn basic network monitoring techniques.
 - Gain practical exposure to cyber security tools and libraries.
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3. Tools and Technologies Used

Tool / Technology	Purpose
Python	Programming Language
Scapy	Packet Capturing and Analysis
Visual Studio Code	Code Editor
GitHub	Version Control and Project Hosting
Npcap	Packet Capture Driver for Windows

4. Project Description

The Basic Network Sniffer is a Python-based application that captures live packets from the network interface.

The application uses the Scapy library to sniff packets and extract important network information. The program displays the Source IP Address, Destination IP Address, and Protocol Number.

For demonstration purposes, the sniffer captures a limited number of packets.

5. Working Principle

The working of the Network Sniffer is based on packet capturing.

Steps Involved

1. The program starts packet sniffing using Scapy.
 2. Network packets flowing through the system are captured.
 3. Each packet is checked for the IP layer.
 4. The Source IP Address, Destination IP Address, and Protocol details are extracted.
 5. The captured information is displayed in the terminal.
 6. The sniffer continues monitoring packets until the specified packet count is reached.
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6. Source Code

```
from scapy.all import sniff, IP

def packet_callback(packet):
    if packet.haslayer(IP):
        print(f"""
        Source IP: {packet[IP].src}
        Destination IP: {packet[IP].dst}
        Protocol: {packet[IP].proto}
        """)

print("Starting Network Sniffer...")
sniff(prn=packet_callback, count=10)
```

Explanation:

The program imports the required Scapy modules, defines a callback function to extract IP packet information, and starts packet sniffing with a limit of 10 packets.

7. Output

When the script is executed, it successfully initializes and begins capturing network packets.

Example Output

```
Starting Network Sniffer...  
  
Source IP: 172.64.148.235  
Destination IP: 192.168.100.8  
Protocol: 6
```

8. Advantages

- Helps understand network traffic.
- Useful for learning cyber security concepts.
- Simple and lightweight implementation.
- Demonstrates packet analysis techniques.
- Useful for network troubleshooting.

9. Limitations

- Captures limited packet information.
- Requires Administrator privileges.
- Works only on local network traffic.
- Cannot decrypt encrypted traffic.

10. Conclusion

This project provided practical experience in packet sniffing and network monitoring using Python and the Scapy library.

It helped in understanding how network packets travel across a network and how security professionals analyze network traffic for monitoring and troubleshooting.

Overall, this project strengthened the understanding of cyber security fundamentals, packet analysis techniques, and practical implementation of network monitoring using Python.